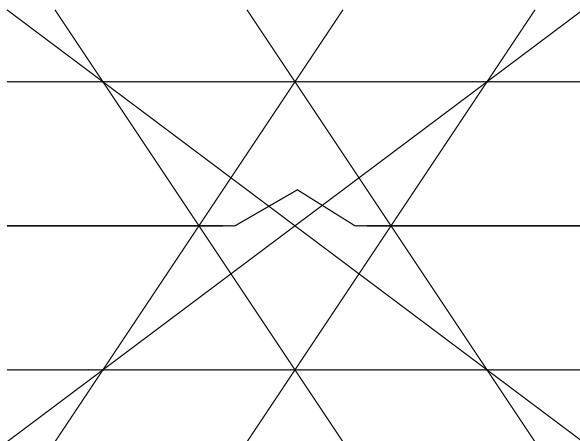


arrangement, the *non-Pappus configuration*.



However, this pseudoline arrangement *does* give an oriented matroid via Theorem 7.26 (uniquely, if we number the pseudolines and specify a positive side for each of them), the *non-Pappus oriented matroid* $\mathcal{M}_{nP}$. This is an oriented matroid of rank 3 on 10 elements. However, if we delete the element that corresponds to the line at infinity, then we get an oriented matroid $\mathcal{M}'_{nP}$ on 9 elements which is nonrealizable as well, since the line at infinity was not needed for the nonrealizability argument.

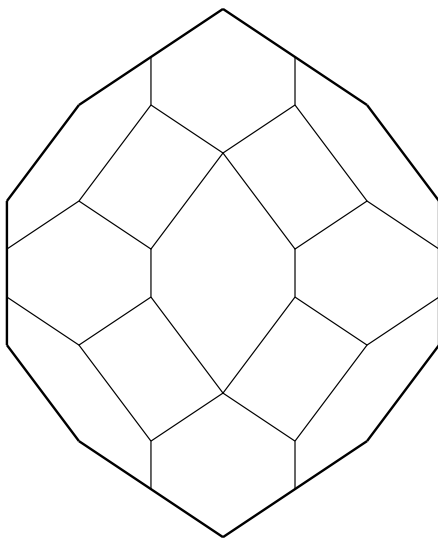
There are even nonrealizable pseudoline arrangements that are simple (no three pseudolines cross in a point or are parallel) — see Ringel [463] [255, p. 42] and Exercise 7.16 for an example with only 9 pseudolines in the plane. In fact, all (simple or nonsimple) arrangements of pseudolines with at most 8 pseudolines are realizable (for this count the line at infinity, if it is there), and Ringel’s example is essentially the unique simple one with 9 pseudolines, according to Richter [455].

7.5 Zonotopal Tilings

What do you “see” if you “look at” a d -dimensional zonotope? You see its “front facets,” which are $(d - 1)$ -dimensional facets and which fill a shape that is a projection of a zonotope, and thus is a zonotope itself.

If $d = 3$ (where this is most likely to happen to you anyway), and you look from a point very far away, the picture might look like the drawing on the next page. Here the shape you see is a 10-gon, filled “face to face” by quadrilaterals and hexagons.

In general, the shape you see is a (parallel) projection of a zonotope, which is a zonotope itself. It is covered by the images of the front facets of the zonotope, which are also zonotopes.



Thus, “looking at zonotopes” leads to zonotopal tilings, which one can formally define as follows.

Definition 7.29. A *zonotopal tiling* of rank d is a $(d - 1)$ -dimensional polyhedral complex $\mathcal{C}$ such that both the union $|\mathcal{C}|$ and the faces $F \in \mathcal{C}$ are zonotopes.

The zonotopal tiling is *regular* if it arises from “viewing” a d -dimensional zonotope from a point at infinity, that is, if it arises from a projection $\pi : Z \longrightarrow |\mathcal{C}|$ of a zonotope Z via the construction of Definition 5.3.

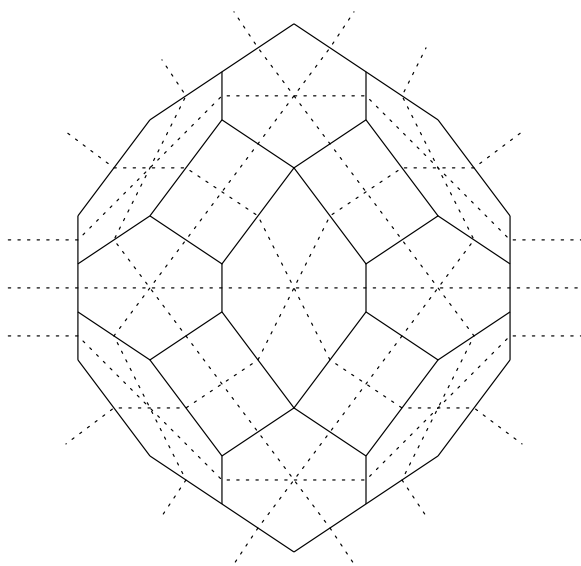
Now if this zonotopal tiling is regular, then it encodes a d -dimensional zonotope, and thus it corresponds to an oriented matroid of rank d .

However, not all zonotopal tilings are regular. To see this, first observe that in the case $d = 3$, there is an obvious way to “draw” an arrangement of pseudolines into the zonotopal tiling, as on the next page.

Note that there is no similar (systematic) way to draw an arrangement of straight lines into the picture — although this is the picture of an actual 3-dimensional zonotope! So we see arrangements of pseudolines, in the polygonal version of Definition 7.25, coming up quite naturally.

Do you recognize the pseudoline arrangement we have just drawn? It is combinatorially equivalent to the Pappus line arrangement that we had constructed before, with an extra horizontal line added through the three special points that have to be collinear anyway, because of Pappus’ theorem. Thus this pseudoline arrangement is certainly realizable!

Lemma 7.30. *All the pseudoline arrangements that come from a regular zonotopal tiling of rank 3 are realizable.*



Proof. Assume (without loss of generality) that the projected 3-zonotope has a “special” summand in the direction of projection. Consider the associated hyperplane arrangement, which contains a dual to the special summand.

Now construct the affine arrangement by intersecting with an affine plane that is parallel to the special one. The resulting arrangement of straight lines is combinatorially equivalent to our pseudoline arrangement; they have the same oriented matroid by construction. $\square$

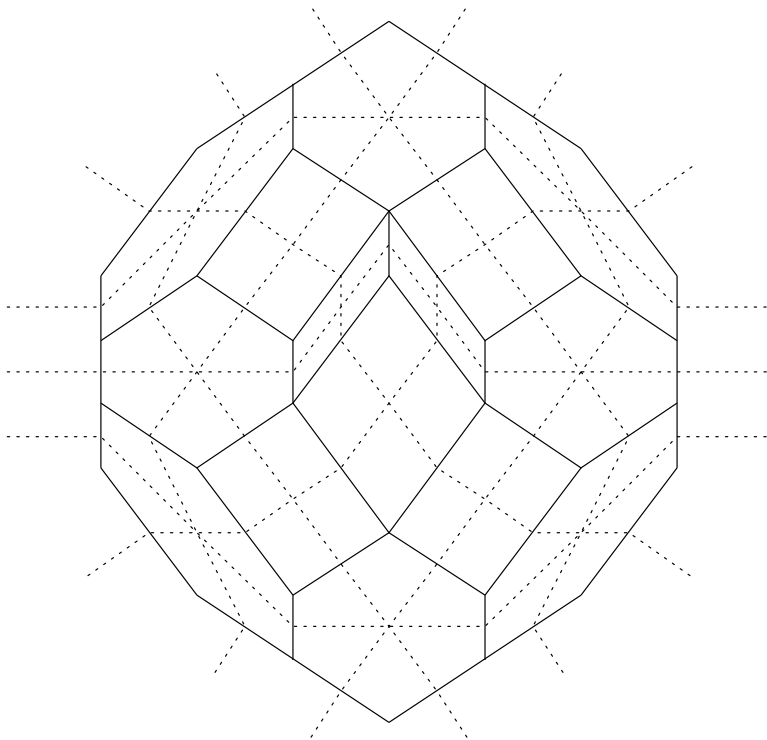
For the zonotopal tiling above, the construction in the proof of Lemma 7.30 produces the line arrangement we had used to illustrate Pappus’ theorem, with the horizontal line through the three dots added.

Now let’s modify the zonotopal tiling just a little, with all the worst intentions, so this will also modify the pseudoline arrangement. We obtain a zonotopal tiling, depicted on the next page, which “realizes” the non-Pappus pseudoline arrangement, which is nonrealizable, inside a zonotopal tiling! What happened?

We are about to hit a quite remarkable result, the Bohné-Dress theorem. It claims that every zonotopal tiling of rank d represents an oriented matroid of rank d , and conversely it characterizes the oriented matroids that can be represented by tilings of a given zonotope Z .

Now it is difficult here to supply the details for the two-way path from zonotopal tilings to pseudoarrangements, which the pictures suggest, since we have not even defined pseudoarrangements and do not intend to do this here (in order to avoid undue topological subtleties). However, everything

we discuss here has a higher-dimensional version, and we want to supply at least the basic tools to describe them.



Thus, we work in a different direction: in view of the topological representation theorem it is sufficient to construct the oriented matroid associated with a zonotopal tiling — and this is just a system of sign vectors, without any topology! Here we go.

The following provides the basic construction and shows how the faces in a zonotopal tiling get sign vectors associated with them, almost canonically.

Construction (with Definitions) 7.31. ([111], [461, Sect. 1])

Let $\mathcal{Z}$ be a zonotopal tiling in $\mathbb{R}^d$.

Two edges $e, e' \in \mathcal{Z}$ are defined to be *equivalent* if there is a sequence $e = e_0, e_1, \dots, e_t = e'$ of edges in $\mathcal{Z}$ such that e_{i-1} and e_i are opposite edges in a 2-face of $\mathcal{Z}$, for $1 \leq i \leq t$.

If this divides the edges in $\mathcal{Z}$ into n equivalence classes, then n is the number of zones of $\mathcal{Z}$. Let $\mathcal{E}_1, \mathcal{E}_2, \dots, \mathcal{E}_n$ enumerate the corresponding equivalence classes of edges. The i th zone of $\mathcal{Z}$, denoted $\mathcal{Z}_i$, is the collection of all those $F \in \mathcal{Z}$ which have a face in $\mathcal{E}_i$.

The edges of an equivalence class are all translates of each other, so we can choose vectors $\mathbf{v}_i \in \mathbb{R}^d$ such that the edges in $\mathcal{E}_i$ are translates of the edge $[-\mathbf{v}_i, \mathbf{v}_i] \subseteq \mathbb{R}^d$. In this situation, we say that the vector configura-

tion $V := (\mathbf{v}_1, \dots, \mathbf{v}_n) \in \mathbb{R}^{n \times d}$ corresponds to the tiling $\mathcal{Z}$. This vector configuration is unique, up to relabeling and to reversal of signs.

The vector configuration V is a multiset: it may contain parallel or antiparallel vectors. It is not hard to see that the zonotope $Z(V)$ it generates is (a translate of) $|\mathcal{Z}|$.

The choice of a vector $\mathbf{v}_i$ also determines a *positive side* and a *negative side* of the zone $\mathcal{Z}_i$. Thus we can associate a sign vector $X^F \in \{+, -, 0\}^n$ with every face $F \in \mathcal{Z}$, via

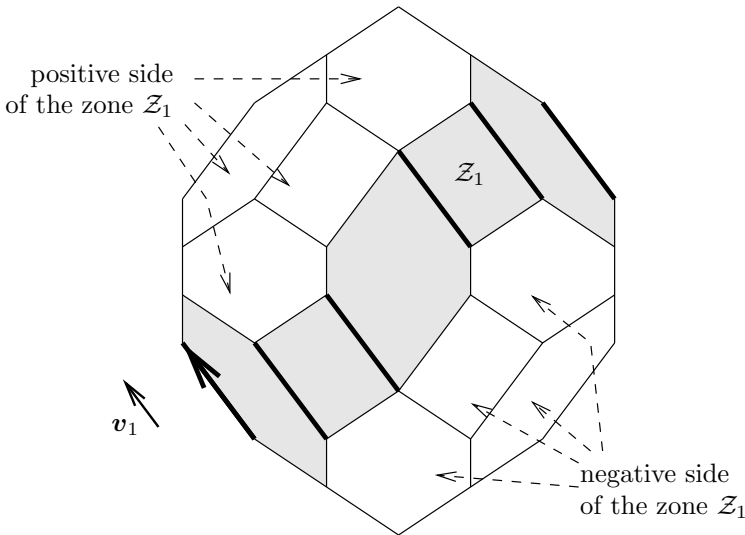
$$X_i^F = \begin{cases} + & \text{if } F \text{ is on the positive side of the zone } \mathcal{Z}_i, \\ 0 & \text{if } F \in \mathcal{Z}_i, \\ - & \text{if } F \text{ is on the negative side of the zone } \mathcal{Z}_i. \end{cases}$$

The set

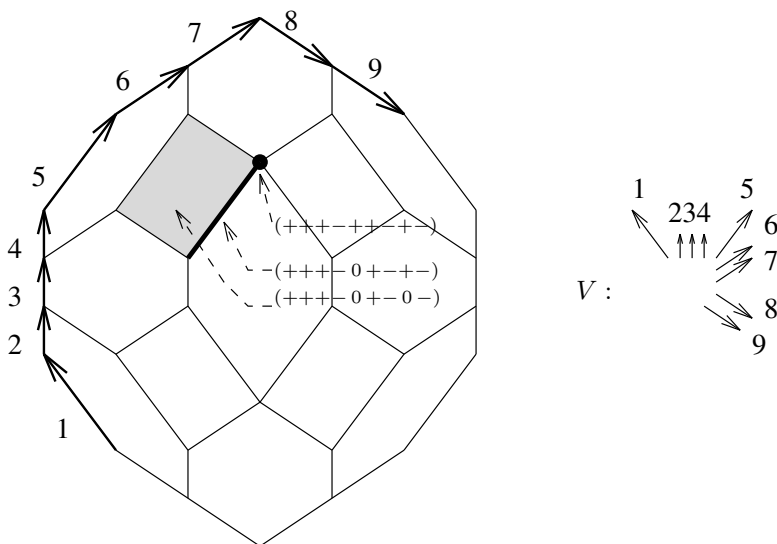
$$\mathcal{O}(\mathcal{Z}) := \{X^F : F \in \mathcal{Z}\}$$

is the family of *affine sign vectors* of $\mathcal{Z}$.

The following sketch shows one zone $\mathcal{Z}_1$ in the Pappus zonotopal tiling. The zone consists of the bold edges (they form the set $\mathcal{E}_1$) and the shaded 2-faces. One possible vector $\mathbf{v}_1$ is indicated, and both the positive and the negative sides depend on this choice: if we replace $\mathbf{v}_1$ by $-\mathbf{v}_1$, then the sides of $\mathcal{Z}_1$ are exchanged.



In the next sketch, all zones have been labeled and directed. The associated sign vector is indicated for one vertex, one edge, and one 2-face of the tiling.



The Bohné-Dress Theorem 7.32. [111, Thms. 4.1, 4.2] [461, Thm. 1.7] *Let $V \in \mathbb{R}^{d,n}$ be a vector configuration of rank d , let $Z := Z(V)$ be its zonotope, and let $\mathcal{V}^* := \mathcal{V}^*(V)$ be its oriented matroid. If $\mathcal{Z}$ is a zonotopal tiling of Z for which V corresponds to $\mathcal{Z}$, then the family of sign vectors*

$$\begin{aligned} \widehat{\mathcal{V}}^* := & \{ (X, +) : X \in \mathcal{O}(\mathcal{Z}) \} \\ & \cup \{ (Y, 0) : Y \in \mathcal{V}^*(V) \} \\ & \cup \{ (-X, -) : X \in \mathcal{O}(\mathcal{Z}) \}. \end{aligned}$$

is an oriented matroid of rank $d+1$. Furthermore, the construction $\mathcal{Z} \rightarrow \widehat{\mathcal{V}}^$ induces a canonical bijection between*

- the zonotopal tilings $\mathcal{Z}$ of $Z(V)$ with associated vector configuration V , and
- the oriented matroids $\widehat{\mathcal{V}}^* \subseteq \{+, -, 0\}^{n+1}$ with

$$\{X \in \{+, -, 0\}^n : (X, 0) \in \widehat{\mathcal{V}}^*\} = \mathcal{V}^*(V).$$

Proof. This is the correspondence suggested by the pseudoarrangement of hyperplanes one can draw into every zonotopal tiling. We have certainly seen that this is plausible, at least in rank 3.

The actual proof of Theorem 7.32, together with the proof that Construction 7.31 works correctly as claimed, is surprisingly difficult. We refer to Bohné's thesis [111], and to the new proof by Richter-Gebert & Ziegler [461]. $\square$

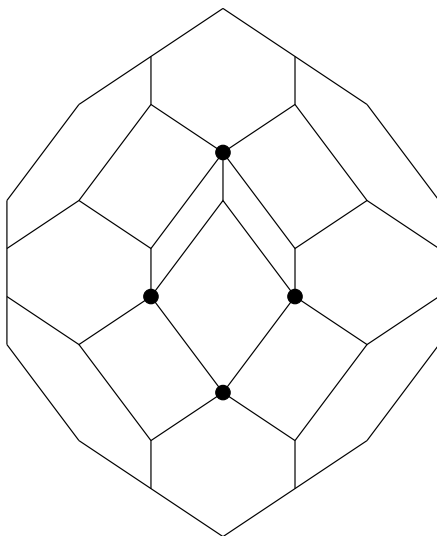
With the Bohne-Dress theorem, we can translate results about oriented matroids into facts about zonotopal tilings, and back. In the following we describe one (striking) instance.

Definition 7.33. A vertex in a zonotopal tiling $\mathcal{Z}$ of rank d is *simple* if it has degree $d - 1$ and is a vertex of $Z = |\mathcal{Z}|$, or if it has degree d and is not a vertex of Z .

From Shannon's Theorem 7.20 we get the following estimate.

Corollary 7.34. Every regular zonotopal tiling of rank d with $n - 1$ zones has at least d simple vertices on the boundary of $Z := |\mathcal{Z}|$, and at least $n - d$ simple vertices in the interior of Z .

One can see by elementary arguments that every zonotopal tiling of rank 3 satisfies these estimates, even if it is not regular (cf. Exercise 7.15). For example, the nonsimple vertices of the non-Pappus zonotopal tiling are marked in the following figure — all the other vertices are simple.



However, in the framework of pseudoarrangements one can construct oriented matroids (in terms of arrangements of pseudohyperplanes) that have fewer than $2n$ simplicial regions. The first example of this kind was presented by Roudneff & Sturmfels [467].

The current “world records” about simple vertices are due to Richter-Gebert [456, Thm. 2.2], who constructed oriented matroids of rank 4 on $4n$ elements that have only $6n$ simplicial regions. Furthermore, from Richter-Gebert's example [456, Thm. 2.3] we get an oriented matroid $R(20)$ of rank 4 which has a pseudohyperplane “8” that is not adjacent to any simplicial region. Furthermore, the restriction to the pseudohyperplane 8 (the contraction $R(20)/8$ of the oriented matroid) is realizable. Via the Bohne-Dress theorem, these results translate into the following theorem.

Theorem 7.35.

- (i) [467] *There is a zonotopal tiling of rank 4 (in $\mathbb{R}^3$), with 7 zones ($n = 8$), which has only 7 simple vertices, and only 3 of them are on the boundary.*
- (ii) [456, Thm. 2.2] *For $k \geq 2$ there are zonotopal tilings of rank 4 (in $\mathbb{R}^3$), with $4k - 1$ zones ($n = 4k$), which have only $3k + 1 = \frac{3}{4}n + 1$ simple vertices.*
- (iii) [456, Thm. 2.3] *There is a 3-dimensional zonotopal tiling with 19 zones, which has no simple vertex on the boundary.*

You should try to visualize these — in view of the “geometric” description of the pseudoplane arrangements in Richter-Gebert’s paper with many drawings this is not out of reach. A photo of a geometric model for the oriented matroid of part (i), built by Bokowski and Richter-Gebert, can be found in [112, p. 562].

Notes

The permutahedron was first written about by Schoute in 1911 [481], it seems. General zonotopes were known to Blaschke [101, p. 250]. The first systematic investigation of zonotopes was in Bolker [119], followed immediately by Schneider [475], and then by McMullen [392], who developed *zonal diagrams* — a version of Gale diagrams suitable for “zonotopes with few summands” (see Exercise 7.7). There is a revived interest now, due to the connection to oriented matroids, hyperplane arrangements, aspects of optimization, computational geometry and convexity, and so on. We refer to the surveys by McMullen [394] and by Schneider & Weil [478], to [96, Sect. 2.2], and to the paper by Gritzmann & Sturmfels [244] and the references therein. More on Minkowski sums can also be found in [244].

The subject of hyperplane arrangements has a lot of different aspects, and we do not even try to give an introduction here. We refer to [96] for the case of real hyperplane arrangements and their oriented matroids, and for further references. Arrangements of lines and pseudolines (corresponding to arrangements of rank 3) are beautifully discussed by Grünbaum in [255].

Fans, polytopal or not, are of great interest for algebraic geometry. In particular they represent toric varieties. In this case, the interest is restricted to fans that are pointed (i.e., $\{\mathbf{0}\} \in \mathcal{F}$) and rational (every cone is generated by rational vectors). We refer to books by Fulton [215] and Oda [426], and in particular to the combinatorial treatment in Ewald [201].

Simple zonotopes exist, but they are rare. As we have seen, they correspond to simplicial arrangements of hyperplanes. Examples of such arrangements arise naturally in the theory of Coxeter groups, root systems,